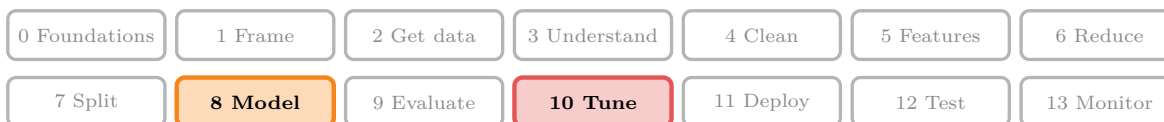


K-Means Clustering: How It Works

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Where this fits: Pipeline map steps 8 (Model), 10 (Tune). See the [Course map](#).



- **Builds on:** Unsupervised learning ([Note 3](#)); Feature scaling ([Note 24](#)).
- **Leads to:** Hierarchical clustering ([Note 131](#)); DBSCAN ([Note 132](#)).
- **Compare with:** Hierarchical clustering ([Note 131](#)); DBSCAN ([Note 132](#)); Gaussian mixture model (GMM) ([Note 640](#)).

1. Overview

Key point: k-means splits the data into k clusters by repeating two moves: give every point to its nearest centroid, then move every centroid to the mean of its points. The elbow method helps us choose k.

K-means (G-996) is a **clustering** (G-401) algorithm: it puts similar **observations** (G-1374; records, one row of the data table each) in the same group without being told what the groups are (clustering is **unsupervised learning**, G-2058: the [types of ML Note](#), section 3.2). Clustering is used everywhere: e-commerce sites cluster customers, colleges cluster students, photo apps cluster images.

The binning Note ([Note 32](#), section 7.1) already ran the k-means loop on a single column. This Note runs it on two **features** (G-772; input variables, one column of the data table each), where each observation is a point and distances are measured in a plane, and adds what was missing there: the starting centroids, the stopping test and how to choose k. Figure 1 shows the whole algorithm on 18 students.

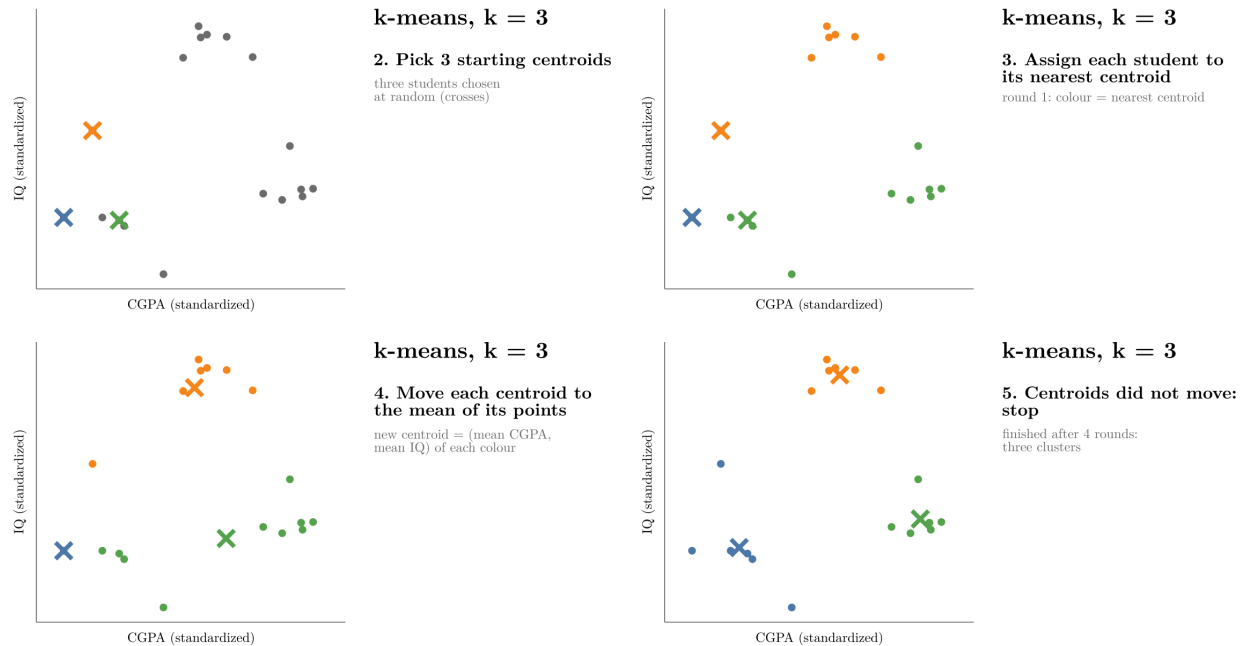


Figure 1: k-means with $k = 3$ on 18 students: random start, assign, move, repeat until the centroids stop moving

2. Prerequisites

- Clustering and unsupervised learning: the [types of ML Note](#).
- k-means on one column, and the word centroid: the [binning Note](#).
- The Euclidean distance: the [KNN imputer Note](#), section 4.1.
- Putting features on one scale (standardization): the [standardization Note](#).

3. Why we need a clustering algorithm

Key point: With two features we can see the clusters by eye; with ten or a hundred features we cannot, and k-means does the same job in any number of dimensions.

Take the training and placement cell of a college. It has the CGPA and IQ of every final-year student and wants to split the students into groups, so that each group can be prepared for placements in its own way.

With only two features, a scatter plot is enough: we look at it and see, say, three groups. No algorithm is needed. Now add marks in class 10, marks in class 12, home state and parents' monthly income: 10 to 15 features. We can no longer draw the data, so we can no longer see the groups.

Many features are where k-means earns its place. We explain k-means in two dimensions, because two dimensions can be drawn. k-means needs only distances and means, and both work with any number of features, exactly as for KNN (the [KNN Note](#), section 2.2), so every step runs unchanged in 100 dimensions.

4. The five steps of k-means

Key point: 1 choose k ; 2 pick k starting centroids; 3 assign every point to its nearest centroid; 4 move each centroid to the mean of its points; 5 stop if no centroid moved, otherwise go back to step 3.

Steps 3 to 5 are the loop of the [binning Note](#), section 7.1: assign every value to its nearest centroid, move each centroid to the mean of its values, repeat until nothing changes. Here we add what that Note left out: choosing k , the starting centroids, and the loop on two features, run on the 18 students of Figure 1.

4.1 Step 1: choose k

Key point: k-means cannot find the number of clusters by itself; we must tell it k .

k is the number of clusters we want. k-means does not work it out from the data: looking at the students, it cannot decide whether there should be 3 or 4 groups. Choosing k is our job.

The requirement sounds odd: if we cannot plot data with many features, how can we know k ? Section 5 answers this with the elbow method. Until then we assume we know it: $k = 3$ for the students.

4.2 Step 2: pick the starting centroids

Key point: k-means picks k points at random and treats them as the first centres of the clusters.

At the start there are no clusters yet, so k-means simply picks k data points at random and calls them **centroids** (G-367). This step is the **centroid initialization** (G-366).

In Figure 1 (top left), three students were picked at random. They are marked with crosses: blue, orange and green. Two of them happen to sit in the same group of students, a poor start; the next steps repair it.

4.3 Steps 3 and 4 on two features

Key point: Distances become Euclidean distances in the plane, and each new centroid is the mean of each feature over the cluster's points.

Assign. For every point we compute its **Euclidean distance** (G-715; the [KNN imputer Note](#), section 4.1) to each of the k centroids, and the point joins the nearest one. With 18 students and 3 centroids, that is $18 \times 3 = 54$ distances per round. In Figure 1 (top right), the green centroid is nearest to every student on the right and at the bottom, so all of them turn green; the orange centroid wins the top group; the blue centroid has only itself.

Move. The new centroid of a cluster is the mean CGPA and the mean IQ of its points. For example, a cluster with the points (1, 2), (3, 2) and (2, 5) gets the centroid

$$\left(\frac{1 + 3 + 2}{3}, \frac{2 + 2 + 5}{3} \right) = (2, 3).$$

centroid = (mean of 1, 3, 2; mean of 2, 2, 5) = (2, 3)
WCSS = 2 + 2 + 4 = 8

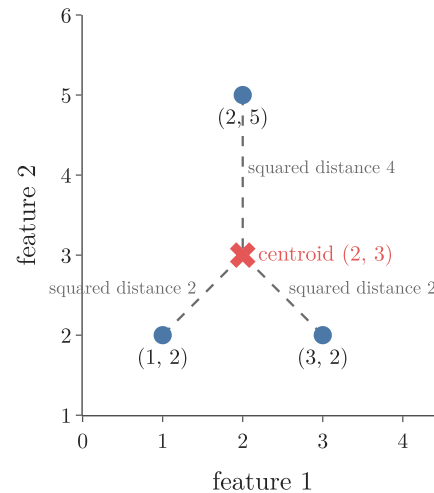


Figure 2: The worked example: the centroid of the three points is the mean of each feature, (2, 3). The dashed lines are the distances used again in section 5.1; their squares add up to the cluster’s WCSS, 8

Figure 2 draws this example.

In Figure 1 (bottom left), the green centroid moves to the middle of its many points, between the right and bottom groups. The orange centroid moves up into the top group.

Extra: CGPA runs from about 4 to 10, IQ from about 70 to 140. In a raw squared Euclidean distance a gap of 10 IQ points adds $10^2 = 100$, while a gap of 2 CGPA points adds only $2^2 = 4$, so IQ would dominate the distances. Like KNN (the [KNN Note](#), section 3.2), k-means is distance-based, so Figure 1 uses standardized values (the [standardization Note](#)). How the features are scaled changes which clusters come out, and standardizing is not always the right choice either (ESL §14.3.3).

4.4 Step 5: stop when the centroids stop moving

Key point: Compare the new centroids with the old ones: if they moved, go back to step 3; if they did not, the clusters are final.

After each move we compare the centroids with where they were in the previous round. If they moved, the clusters may still change, so we assign every point again. If they did not, assigning again would give the same clusters and moving again the same centroids: the algorithm has finished. This is **convergence**.

In Figure 1, round 2 hands most bottom-left students, and the lone orange student on the left, to the blue centroid, which is now nearer to them. Round 3 moves the last one, the student at the very bottom. In round 4 no point changes cluster, so no centroid moves, and three clean clusters remain (bottom right).

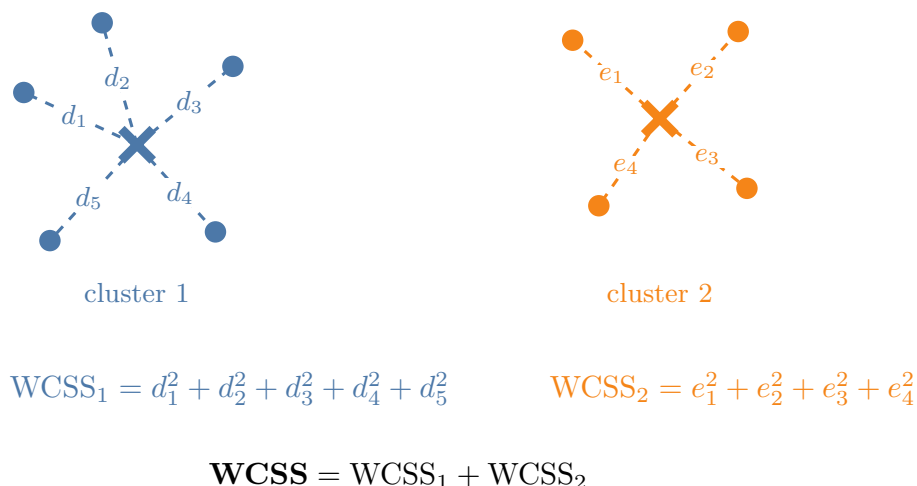
5. Choosing k: the elbow method

Key point: Run k-means for $k = 1, 2, 3, \dots$, plot the WCSS of each, and pick the k where the curve bends from steep to flat: the **elbow method** (G-671).

5.1 WCSS: how tight the clusters are

Key point: WCSS adds up, over all clusters, the squared distance from each point to its own centroid. Small WCSS means tight clusters.

WCSS (G-2102; within-cluster sum of squares) measures how far the points sit from their own centroids. scikit-learn calls the same number **inertia**. Figure 3 shows how it is built.



× = centroid; dashed line = distance from a point to its own centroid

Figure 3: WCSS of one cluster: the squared distances from each point to its centroid, added up; WCSS of several clusters: the sum over the clusters

1. **In words:** for every point, take its distance to the centroid of its own cluster and square it. Add these up within each cluster, then add the clusters' totals.
2. **Formula:** with clusters $C_1, \dots, C_k$ and centroids $c_1, \dots, c_k$,

$$\text{WCSS} = \sum_{j=1}^k \sum_{x \in C_j} \|x - c_j\|^2$$

where $\|x - c_j\|$ is the Euclidean distance from point x to centroid c_j .

3. **Example:** the cluster (1, 2), (3, 2), (2, 5) from section 4.3 has centroid (2, 3). The squared distances are

$$(1 - 2)^2 + (2 - 3)^2 = 2, \quad (3 - 2)^2 + (2 - 3)^2 = 2, \quad (2 - 2)^2 + (5 - 3)^2 = 4,$$

so this cluster's WCSS is $2 + 2 + 4 = 8$. If a second cluster had WCSS 5, the total would be $8 + 5 = 13$.

5.2 The elbow curve

Key point: The elbow curve plots WCSS against the number of clusters, $k = 1$ up to about 10 or 20.

To draw the **elbow curve** (G-670), we run k-means once for each k and record the WCSS:

1. **$k = 1$:** all points form one cluster, its centroid is the mean of all the data, and we compute the WCSS.
2. **$k = 2$:** k-means makes two clusters; we compute WCSS of each and add them.
3. **$k = 3, 4, \dots$:** the same, up to 10, 15 or 20.

Going much beyond 20 is rarely useful. A business will not treat its customers in 50 different ways.

5.3 WCSS always falls as k grows

Key point: More clusters means every point sits closer to a centroid, so WCSS shrinks; with one cluster per point it reaches 0.

Is WCSS for 1 cluster bigger than for 2, and for 2 bigger than for 3? Yes. With more centroids, each point can find one closer to it, so the squared distances shrink.

The extreme case proves the trend. Say there are 50 points. The largest possible k is 50: every point is its own cluster and its own centroid. Every distance from a point to its centroid is 0, so $WCSS = 0$.

So the curve starts high at $k = 1$ and falls towards 0 as k approaches the number of points. Picking the k with the smallest WCSS would therefore always pick the largest k , which is useless. We need a different rule.

5.4 Reading the elbow

Key point: The elbow point is where WCSS stops falling fast: adding one more cluster after it gains little.

Figure 4 is the elbow curve for a real dataset: 272 eruptions of the Old Faithful geyser, each described by two features, the eruption's duration and the waiting time before it (standardized, as section 4.3 recommends). The dataset also labels every eruption as short or long: two kinds. WCSS falls from 544 at $k = 1$ to 80 at $k = 2$. After that it hardly moves: 56 at $k = 3$, 44 at $k = 4$, 34 at $k = 5$.

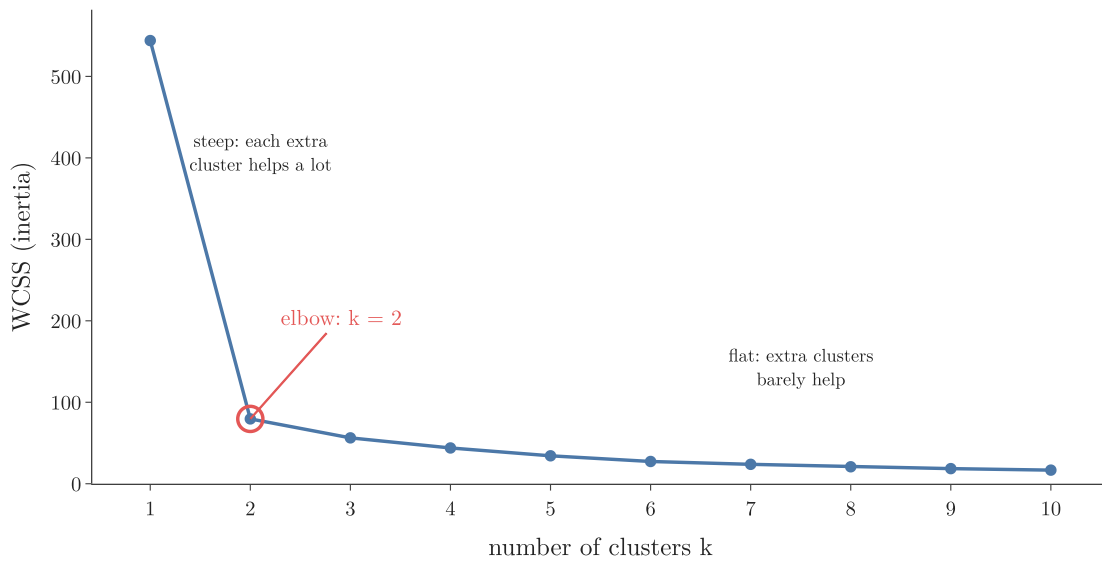


Figure 4: Elbow curve: WCSS against k for the 272 Old Faithful eruptions (standardized); the curve bends at $k = 2$

The **elbow point** (G-672) is where the curve bends, the k after which the fall flattens out. Here it is $k = 2$, the two kinds of eruption:

- Going from 1 to 2 clusters cuts WCSS by 85%: the second cluster was worth it.
- Going from 2 to 3, and beyond, barely helps: the extra clusters only split real groups in pieces (Figure 5).

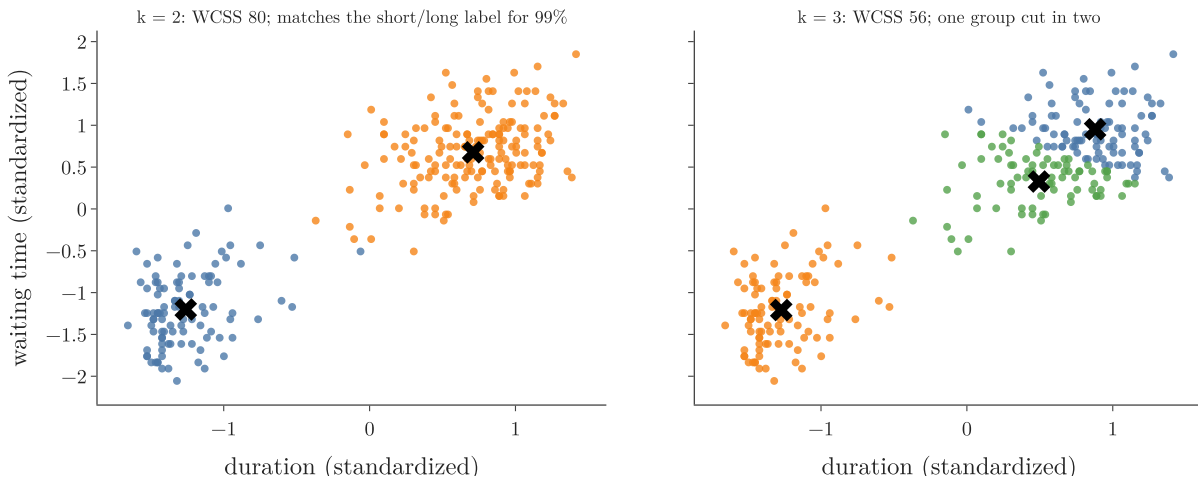


Figure 5: k -means on the 272 eruptions with $k = 2$ (left), the elbow, and $k = 3$ (right). The two clusters at $k = 2$ agree with the dataset's short and long label for 99% of the eruptions; $k = 3$ cuts the long eruptions in two

A memorable picture: the curve is a hill we slide down from the left. On the steep part we drop fast. The point where the slope suddenly eases is the elbow: from there on, each extra cluster buys almost no drop.

Extra: On real data the bend is often not sharp, and it is easy to read the wrong k from it (Schubert 2022). Other ways to choose k exist for this reason, such as the silhouette score (Rousseeuw 1987).

6. Summary

Step	What happens	With the students ($k = 3$)
1. Choose k	We decide the number of clusters	$k = 3$, later checked by the elbow method
2. Initialize	k random points become the centroids	three students picked at random
3. Assign	Every point joins its nearest centroid (Euclidean distance)	$18 \times 3 = 54$ distances per round
4. Move	Each centroid moves to the mean of its points	mean CGPA, mean IQ per cluster
5. Check	Centroids unchanged: stop; else back to step 3	stopped after 4 rounds

- k -means only needs distances and means, so it works the same in 2 or 100 dimensions.
- Scale the features first: k -means is distance-based.
- WCSS (inertia) = the sum of squared distances from each point to its own centroid.
- WCSS always falls as k grows and reaches 0 when every point is its own cluster.
- The elbow method picks the k where the WCSS curve bends from steep to flat.

7. Sources

Built from

- CampusX, “K-Means Clustering Algorithm | Geometric Intuition | Clustering | Unsupervised Learning”, YouTube, <https://www.youtube.com/watch?v=5shTLzwAdEc>

Other references

- ESL: Hastie, Tibshirani and Friedman, *The Elements of Statistical Learning*, 2nd ed., Springer, 2009, §14.3.3 (object dissimilarity and standardization in clustering).
- Schubert 2022: E. Schubert, *Stop using the elbow criterion for k -means and how to choose the number of clusters instead*, arXiv:2212.12189, 2022.
- Rousseeuw 1987: P. J. Rousseeuw, *Silhouettes: a graphical aid to the interpretation and validation of cluster analysis*, Journal of Computational and Applied Mathematics 20, 1987, 53–65.

8. Key terms

Term	Meaning
k	The number of clusters k -means makes; chosen by us
Centroid initialization	Picking the first k centroids, here at random from the data

Term	Meaning
Convergence (k-means)	The point where the centroids stop moving between rounds, so the algorithm stops
WCSS (inertia)	Within-cluster sum of squares: the sum of squared distances from each point to its own centroid
Elbow curve	A plot of WCSS against the number of clusters k
Elbow method	Choosing k at the point where the elbow curve bends from steep to flat
Elbow point	The k after which adding clusters barely lowers WCSS